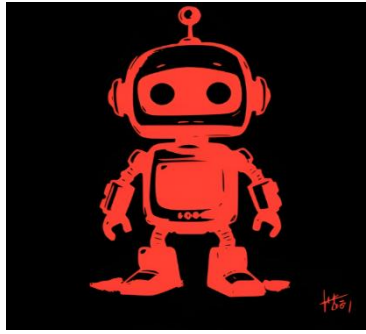




# AEGIS

## The Alchemists

| Alchemist       | Student Number |
|-----------------|----------------|
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## USER MANUAL

**COS301 AEGIS TEAM**  
UNIVERSITY OF PRETORIA  
PRETORIA 012

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# INTRODUCTION

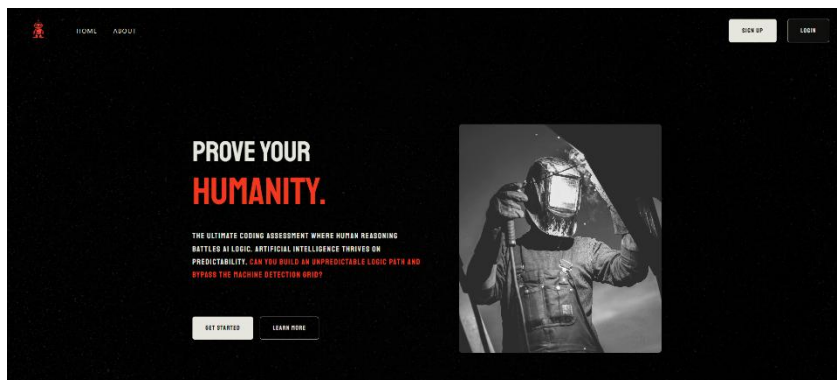
Welcome to AEGIS, an advanced technical assessment platform developed by ‘The Alchemists’ (COS301 AEGIS Team, University of Pretoria). AEGIS evaluates true human reasoning by combining technical challenges with integrity analytics.

## USER GROUPS

## GETTING STARTED:

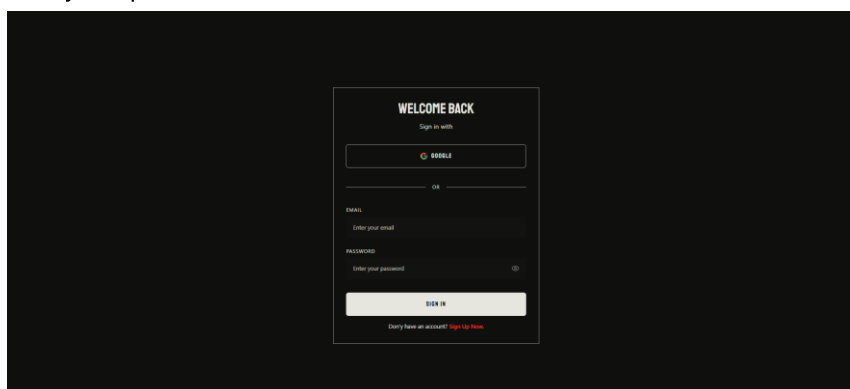
### LANDING PAGE

The landing page tells you about AEGIS. Shows the main way to get into the platform. Use the navigation or the sign-in action to move on to the application.



### SIGNING IN

The sign-in page is used to check who a user is. Enter the email address you signed up with and your password, then click the button to submit the information.



# RECRUITER WORKFLOW

Recruiter uses AEGIS to manage assessments. Recruiter responsibilities are. Include creating, deleting, and assigning assessments.

## MAIN NAVIGATION

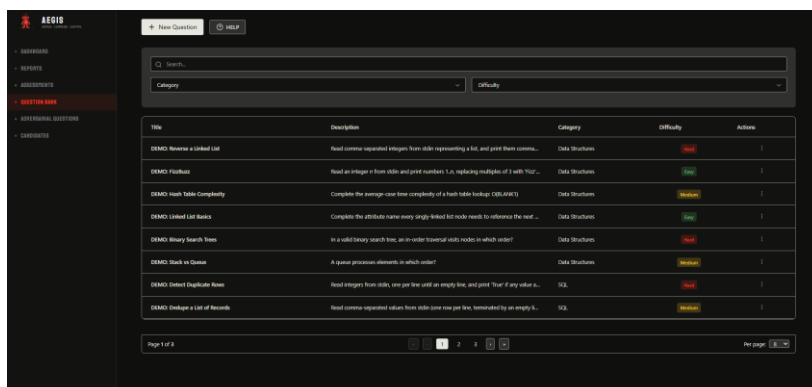
- **Dashboard:** Provides a high-level view of assessment activity and system signals.
- **Reports:** Provides assessment and candidate performance information.
- **Assessments:** The area used to manage assessments.
- **Question Bank:** The question-management area shown in the application navigation.
- **Adversarial Questions:** The area shown for adversarial-question content.
- **Candidates:** The area shown in the application navigation.

## QUESTION BANK

A core feature of AEGIS is the creation and maintenance of technical problems through two specialized modules in the navigation panel:

### STANDARD QUESTION BANK

Serves as the central repository for standard technical coding questions across domains such as Data Structures, Algorithms, Python, and SQL.



The screenshot shows the AEGIS application interface. On the left is a navigation sidebar with links for Dashboard, Reports, Assessments, Question Bank (highlighted), Adversarial Questions, and Candidates. The main content area is titled 'New Question' and 'HELP'. It features a search bar and a filter dropdown for 'Category'. Below this is a table of questions.

| Title                          | Description                                                                                     | Category        | Difficulty | Actions |
|--------------------------------|-------------------------------------------------------------------------------------------------|-----------------|------------|---------|
| DEMO: Reverse a Linked List    | Read common sequential integers from slide representing a list, and print them comma...         | Data Structures | Easy       | ?       |
| DEMO: Fibonacci                | Read an integer n from slide and print number's n, replacing multiples of 3 with "fib..."       | Data Structures | Easy       | ?       |
| DEMO: Hash Table Complexity    | Complete the average-case time complexity of a hash table lookup: O(1/ K )                      | Data Structures | Medium     | ?       |
| DEMO: Linked List Basics       | Complete the attribute name every singly-linked list node needs to reference the next...        | Data Structures | Easy       | ?       |
| DEMO: Binary Search Tree       | In a valid binary search tree, an in-order traversal visits nodes in which order?               | Data Structures | Easy       | ?       |
| DEMO: Stack vs Queue           | A queue processes elements in which order?                                                      | Data Structures | Medium     | ?       |
| DEMO: Detect Duplicate Rows    | Read integers from slide, one per line until an empty line, and print "true" if any value is... | SQL             | Easy       | ?       |
| DEMO: Dedupe a List of Records | Read common sequential values from slide (one row per line, terminated by an empty li...        | SQL             | Medium     | ?       |

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Per page: 10

## STANDARD QUESTION CREATION

Define titles, detailed problem prompts, input/output constraints, sample test cases, and hidden unit tests for auto-grading.

Assign specific categories (e.g., Data Structures, Python, SQL) and difficulty tiers (Easy, Medium, Hard) to ensure balanced test distribution. Click ‘Save’ when ready to deploy the question

CREATE QUESTION

Configure your assessment question.

QUESTION METADATA

QUESTION TITLE

QUESTION COMMENT

DESCRIPTION

CATEGORY

Select Category...

DIFFICULTY

EASY

MEDIUM

HARD

SCORE ALLOCATION

10

MOCKED TEST PROMPT - OR LINKER TO ADD A TOOL

Type a tag

CANCEL

Save

Difficulty

| Category                                                            | Difficulty | Actions |
|---------------------------------------------------------------------|------------|---------|
| join from table representing a list, and print them comma...        | Easy       |         |
| and print numbers 1..N, replacing multiples of 3 with 'fizz'...     | Easy       |         |
| time complexity of a hash table lookup (O(1) time)                  | Medium     |         |
| every singly-linked list needs to reference the next ...            | Easy       |         |
| an in-order traversal visits nodes in which order?                  | Easy       |         |
| in which order?                                                     | Medium     |         |
| 2 per line until an empty line, and print 'fizz' if any value is... | Easy       |         |
| as from table rows row per line, terminated by an empty li...       | Easy       |         |

1

2

3

4

5

Per page: 10

CREATE QUESTION

Configure your assessment question.

FUNCTION SIGNATURE

def

foo(x: int, y: int) -> int

DESCRIPTION

SOLUTION ARCHITECTURE

TEST CASES

Python Literal, Input

1, 2, 3

CANCEL

Save

Difficulty

| Category                                                            | Difficulty | Actions |
|---------------------------------------------------------------------|------------|---------|
| join from table representing a list, and print them comma...        | Easy       |         |
| and print numbers 1..N, replacing multiples of 3 with 'fizz'...     | Easy       |         |
| time complexity of a hash table lookup (O(1) time)                  | Medium     |         |
| every singly-linked list needs to reference the next ...            | Easy       |         |
| an in-order traversal visits nodes in which order?                  | Easy       |         |
| in which order?                                                     | Medium     |         |
| 2 per line until an empty line, and print 'fizz' if any value is... | Easy       |         |
| as from table rows row per line, terminated by an empty li...       | Easy       |         |

1

2

3

4

5

Per page: 10

# ASSESSMENTS

## CREATE ASSESSMENT

Define the assessment. Fill in the required details. Choose the content that should be included in the assessment using the available options.

The screenshot shows the 'CREATE ASSESSMENT' form in the 'ASSESSMENT ANSERAL' section. The form is titled 'CREATE ASSESSMENT' and has a subtitle 'Create an assessment or assessment component'. It features a progress bar with four steps: 'NAME', 'QUESTIONS', 'ANSWERS', and 'EXPORT'. The 'NAME' step is currently active. The form includes the following fields:

- ASSESSMENT IDENTITY**
  - NAME**: A text input field with the placeholder 'Name (Maximum 100 characters)'.
  - DESCRIPTION**: A text input field with the placeholder 'Describe the purpose of the questions'.
  - QUESTIONS COUNT**: A dropdown menu with 'Default' selected.
  - QUESTIONS TYPE**: A dropdown menu with 'MC' selected.
- Time (s)**: A text input field.
- CONTINUE**: A button.

The screenshot shows the 'CREATE ASSESSMENT' form in the 'ASSESSMENT ANSERAL' section. The form is titled 'CREATE ASSESSMENT' and has a subtitle 'Create an assessment or assessment component'. It features a progress bar with four steps: 'NAME', 'QUESTIONS', 'ANSWERS', and 'EXPORT'. The 'QUESTIONS' step is currently active. The form includes the following fields:

- PICK QUESTIONS**
  - Search questions**: A text input field.
  - Filter**: A dropdown menu with 'All questions' selected.
  - QUESTIONS**: A list of questions with checkboxes and expand/collapse icons. The questions are:
    - What is the capital of France?** (MC) [Selected]
    - What is the capital of Germany?** (MC)
    - What is the capital of the United Kingdom?** (MC)
    - What is the capital of Italy?** (MC)
    - What is the capital of Spain?** (MC)
- BACK**: A button.
- CONTINUE**: A button.

The screenshot shows the 'CREATE ASSESSMENT' form in the 'ASSESSMENT ANSERAL' section. The form is titled 'CREATE ASSESSMENT' and has a subtitle 'Create an assessment or assessment component'. It features a progress bar with four steps: 'NAME', 'QUESTIONS', 'ANSWERS', and 'EXPORT'. The 'ANSWERS' step is currently active. The form includes the following fields:

- ANSWER KEY**
  - For the questions identified in the previous step, select the correct answer. If you are unsure, you can select 'None'.**
  - QUESTIONS**: A list of questions with checkboxes and expand/collapse icons. The questions are:
    - What is the capital of France?** (MC) [Selected]
    - What is the capital of Germany?** (MC)
    - What is the capital of the United Kingdom?** (MC)
    - What is the capital of Italy?** (MC)
    - What is the capital of Spain?** (MC)
- BACK**: A button.
- CONTINUE**: A button.

The screenshot shows the 'CREATE ASSESSMENT' form in the 'ASSESSMENT ANSERAL' section. The form is titled 'CREATE ASSESSMENT' and has a subtitle 'Create an assessment or assessment component'. It features a progress bar with four steps: 'NAME', 'QUESTIONS', 'ANSWERS', and 'EXPORT'. The 'EXPORT' step is currently active. The form includes the following fields:

- READY TO GO**
  - File Download**: A text input field.
  - File Upload**: A text input field.
  - Time (s)**: A text input field.
  - Questions & Answers**: A text input field.
  - Questions & Answers**: A text input field.
- BACK**: A button.
- EXPORT ASSESSMENT**: A button.

# CANDIDATE WORKFLOW:

Press Start to open and start the assessment

